Stony Brook Institute at Anhui University

安徽大学纽约石溪学院

# Chapter 5
# Basic Concepts of Statistics

# Common sample distribution

- chi-square distribution
- t distribution
- F distribution
- Quantiles of distribution

# Quantiles

One measure of variation is the vary range, which is the difference between the maximum and the minimum value of random variables.

Quantiles are the tool to describe the vary.

# Upper quantiles of distribution

Definition:

If r.v. X has the distribution F(x) .The constant $\boldsymbol{F_\alpha}$ is upper quantile with the level $\boldsymbol{\alpha}(\mathbf{0} < \boldsymbol{\alpha} < \mathbf{1})$ if

$$P\{X > F_\alpha\} = \alpha$$

This means the probability that r.v. X lies to the right of this value $\boldsymbol{F_\alpha}$ is $\boldsymbol{\alpha}$ .

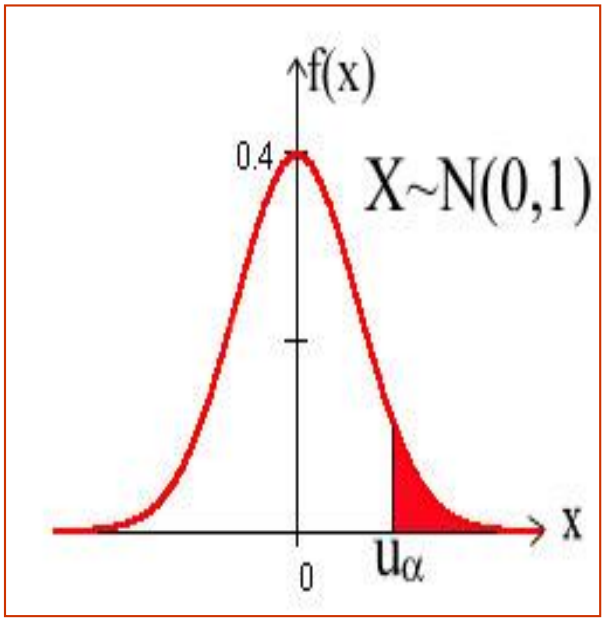
# Upper quantiles of N(0,1) distribution

Definition: The value $u_\alpha$ is the upper α percentile of the N(0,1)distribution, such that α($\mathbf{0} < \boldsymbol{\alpha} < \mathbf{1}$) of the probability (or α of the area) lies to the right of this value. That is

$$P\{X > u_\alpha\} = \frac{1}{\sqrt{2\pi}} \int_{u_\alpha}^{+\infty} e^{-\frac{x^2}{2}} \mathrm{d}x$$

$$= 1 - P\{X \le u_\alpha\} = 1 - \Phi(u_\alpha)$$

$$= \alpha$$



# Upper quantiles of N(0,1) distribution

From the result

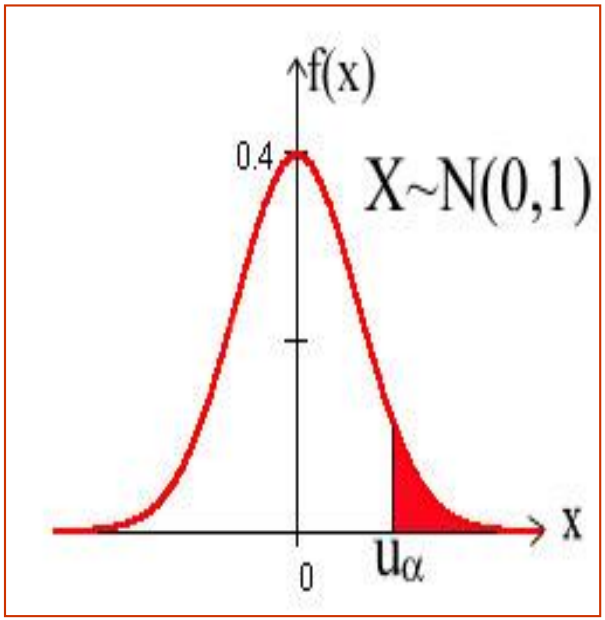
$$\alpha = P\{X > u_\alpha\} = 1 - P\{X \le u_\alpha\} = 1 - \Phi(u_\alpha)$$

we get

a. $\Phi(u_\alpha) = 1 - \alpha$

b. $\boldsymbol{u_{1-\alpha} = -u_\alpha}$.

c. $u_{0.05} = 1.645,\ u_{0.025} = 1.96$



# 1. $\chi^2$ ( chi-square) distribution

1） Definition

If $X_1, X_2, \cdots, X_n \ i.i.d. \sim N(0,\ 1)$, then statistic

$$\chi^2 = X_1^2 + X_2^2 + \cdots + X_n^2$$

has a distribution with degrees of freedom (df) n, which is denoted as

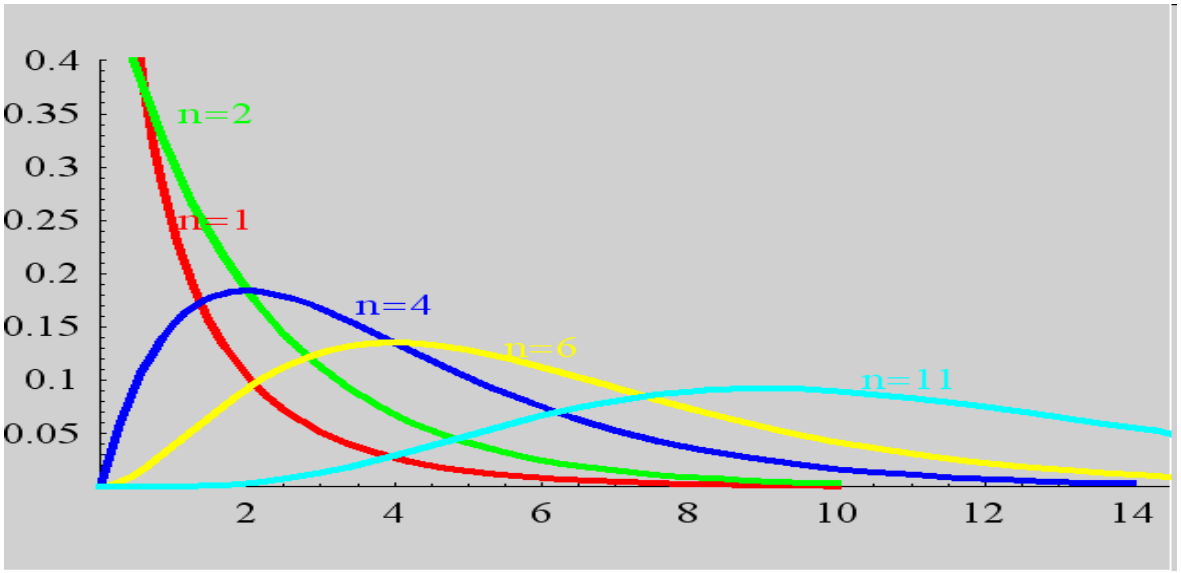
$$\chi^2 \sim \chi^2(n).$$

Denote:

$$\chi^2(n) = \Gamma\left(\frac{n}{2}, \frac{1}{2}\right)$$

2) chi-square distribution's pdf

$$p(x) = \begin{cases} \dfrac{1}{2^{\frac{n}{2}}\Gamma(\frac{n}{2})} x^{\frac{n}{2}-1} e^{-\frac{x}{2}} & x > 0 \\ 0 & else \end{cases}$$



3） The mean and the variance of the chi-square distribution is

$$\mathrm{E}(\chi^2) = n, \qquad \mathrm{Var}(\chi^2) = 2n.$$

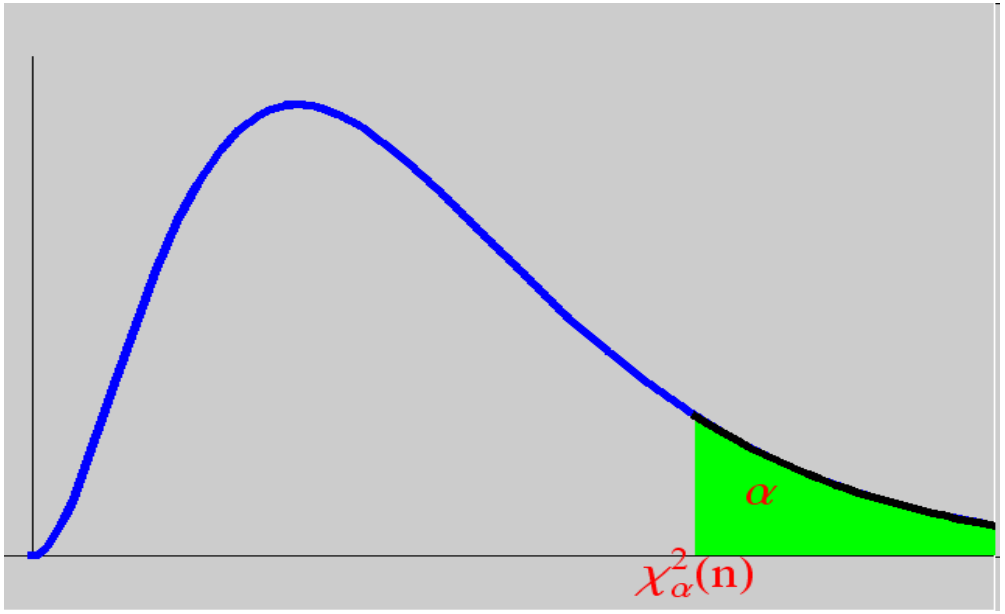
4) If the random variables $X_1, \ldots, X_n$ are independent and with distribution $X_i \sim \chi^2(n_i)$ for i = 1, 2, . . . , m, then

$$\sum_{i=1}^{m} X_i^2 \sim \chi^2(n_1 + n_2 + \cdots + n_m).$$

# Upper quantiles of chi-square distribution

Definition: The value $\chi^2_\alpha(n)$ is the upper α percentile of the chi-square distribution with parameter n , such that α($\mathbf{0} < \boldsymbol{\alpha} < \mathbf{1}$) of the probability (or α of the area) lies to the right of this value. That is

$$P\{\chi^2 > \chi^2_\alpha(n)\} = \alpha$$

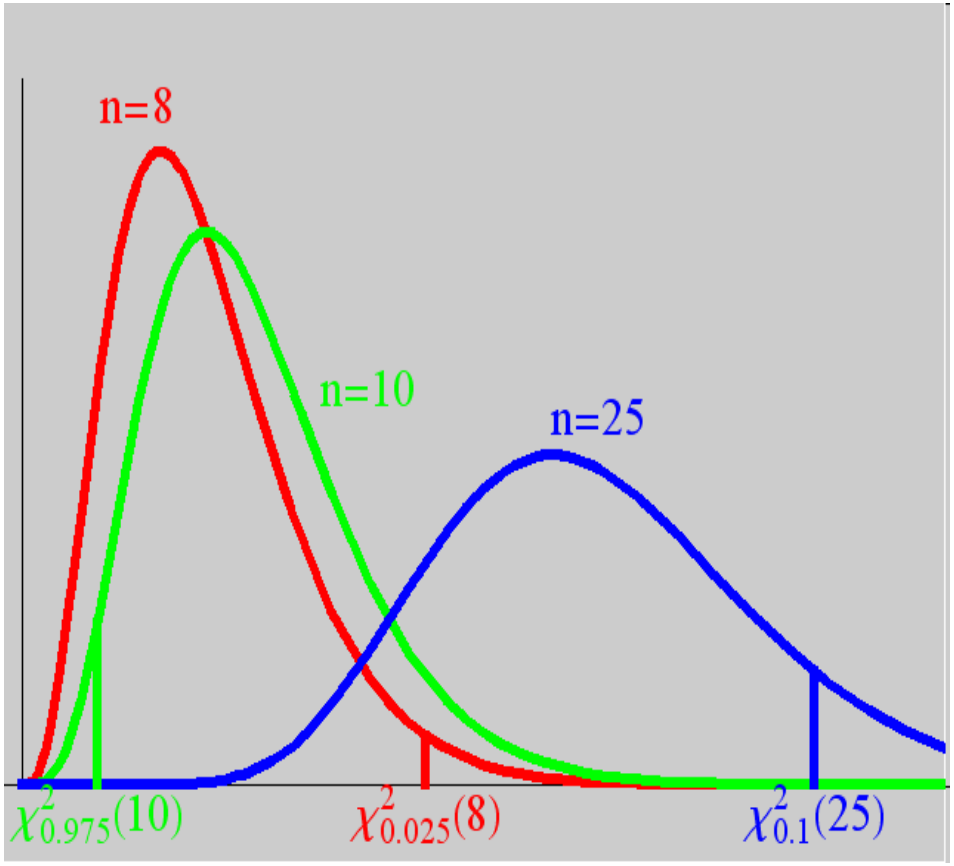


Look up the quantile table of chi-square distribution,

$$\chi^2_{0.025}(8) = 17.535$$

$$\chi^2_{0.975}(10) = 3.247$$

$$\chi^2_{0.1}(25) = 34.382$$



# $\chi^2$ Distribution Table

| $n$ | $\alpha = 0.25$ | 0.10 | 0.05 | 0.025 | 0.01 | 0.005 |
|---|---|---|---|---|---|---|
| 1 | 1.323 | 2.706 | 3.841 | 5.024 | 6.635 | 7.879 |
| 2 | 2.773 | 4.605 | 5.991 | 7.378 | 9.210 | 10.597 |
| 3 | 4.108 | 6.251 | 7.815 | 9.348 | 11.345 | 12.838 |
| 4 | 5.385 | 7.7 | | 11.143 | 13.277 | 14.860 |
| 5 | 6.626 | 9.2 | | 12.833 | 15.086 | 16.750 |
| 6 | 7.841 | 10.6 | | 14.449 | 16.812 | 18.548 |
| 7 | 9.037 | 12.017 | 14.067 | 16.013 | 18.475 | 20.278 |
| 8 | 10.219 | 13.362 | 15.507 | 17.535 | 20.090 | 21.955 |
| 9 | 11.389 | 14.684 | 16.919 | 19.023 | 21.666 | 23.589 |
| 10 | 12.549 | 15.987 | 18.307 | 20.483 | 23.209 | 25.188 |
| 11 | 13.701 | 17.275 | 19.675 | 21.920 | 24.725 | 26.757 |
| 12 | 14.845 | 18.549 | 21.026 | 23.337 | 26.217 | 28.299 |
| 13 | 15.984 | 19.812 | 22.362 | 24.736 | 27.688 | 29.891 |
| 14 | 17.117 | 20.064 | 23.685 | 26.119 | 29.141 | 31.319 |
| 15 | 18.245 | 22.307 | 24.996 | 27.488 | 30.578 | 32.801 |
| 16 | 19.369 | 23.542 | 26.296 | 28.845 | 32.000 | 34.267 |

**17.535**

# $\chi^2$ distribution table

| $n$ | $\alpha$ =0.995 | 0.99 | 0.975 | 0.95 | 0.90 | 0.75 |
|---|---|---|---|---|---|---|
| 1 | – | – | 0.001 | 0.004 | 0.016 | 0.102 |
| 2 | 0.010 | 0.020 | 0.051 | 0.103 | 0.211 | 0.575 |
| 3 | 0.072 | 0.115 | 0.216 | 0.352 | 0.584 | 1.213 |
| 4 | 0.207 | 0.297 | 0.484 | 0.711 | 1.064 | 1.923 |
| 5 | 0.412 | 0.554 | 0.831 | 1.145 | 1.610 | 2.675 |
| 6 | 0.676 | 0.872 | 1.237 | | | 3.455 |
| 7 | 0.989 | 1.239 | 1.690 | | | 4.255 |
| 8 | 1.344 | 1.646 | 2.180 | | | 5.071 |
| 9 | 1.735 | 2.088 | 2.700 | 3.325 | 4.168 | 5.899 |
| 10 | 2.156 | 2.558 | 3.247 | 3.940 | 4.865 | 6.737 |
| 11 | 2.603 | 3.053 | 3.816 | 4.575 | 5.578 | 7.584 |
| 12 | 3.074 | 3.571 | 4.404 | 5.226 | 6.304 | 8.438 |
| 13 | 3.565 | 4.107 | 5.009 | 5.892 | 7.042 | 9.299 |
| 14 | 4.075 | 4.660 | 5.629 | 6.571 | 7.790 | 10.165 |
| 15 | 4.601 | 5.229 | 6.262 | 7.261 | 8.547 | 11.037 |
| 16 | 5.142 | 5.812 | 6.908 | 7.962 | 9.312 | 11.912 |

**3.247**

# $\chi^2$ Distribution Table

| $n$ | $\alpha$ =0.25 | 0.10 | 0.05 | 0.025 | 0.01 | 0.005 |
|---|---|---|---|---|---|---|
| 17 | 20.489 | 24.769 | 27.587 | 30.191 | 33.409 | 35.718 |
| 18 | 21.605 | 25.989 | 28.869 | 31.526 | 34.805 | 37.156 |
| 19 | 22.718 | 27.204 | 30.144 | 32.852 | 36.191 | 38.582 |
| 20 | 23.828 | 28.412 | 31.410 | 34.170 | 37.566 | 39.997 |
| 21 | 24.935 | 29.615 | [illegible] | [illegible]79 | 38.932 | 41.401 |
| 22 | 26.039 | 30.813 | [illegible] | [illegible]81 | 40.289 | 42.796 |
| 23 | 27.141 | 32.007 | [illegible] | [illegible]076 | 41.638 | 44.181 |
| 24 | 28.241 | 33.196 | 36.415 | 39.364 | 42.980 | 45.559 |
| 25 | 29.339 | 34.382 | 37.652 | 40.646 | 44.314 | 46.928 |
| 26 | 30.435 | 35.563 | 38.885 | 41.923 | 45.642 | 48.290 |
| 27 | 31.528 | 36.741 | 40.113 | 43.194 | 46.963 | 49.645 |
| 28 | 32.620 | 37.916 | 41.337 | 44.461 | 48.278 | 50.993 |
| 29 | 33.711 | 39.087 | 42.557 | 45.712 | 49.588 | 52.336 |
| 30 | 34.800 | 40.256 | 43.773 | 46.979 | 50.892 | 53.672 |
| 31 | 35.887 | 41.422 | 44.985 | 48.232 | 52.191 | 55.003 |
| 32 | 36.973 | 42.585 | 46.194 | 49.480 | 53.486 | 56.328 |

**34.382**

$$\chi_{\alpha}^{2}(n) \approx \frac{1}{2}(z_{\alpha} + \sqrt{2n-1})^{2}$$

R.AFisher

**or:**

$$\chi_{\alpha}^{2}(n) \approx n + \sqrt{2n} \cdot z_{\alpha}$$

**eg:**

$$\chi_{0.05}^{2}(50) \approx \frac{1}{2}(z_{0.05} + \sqrt{2 \times 50 - 1})^{2}$$

$$= \frac{1}{2}(1.645 + \sqrt{99})^{2} = 67.221$$

# Ronald Aylmer Fisher

**Great statistican**

**Born: 17 Feb 1890 in London, England**
**Died: 29 July 1962 in Adelaide, Australia**

# 2. t (Student' s t)distribution --- t(n)

1) Definition

If r.v. X,Y are independent, $X \sim N(0,1),\ Y \sim \chi^2(n),$

$$T = \frac{X}{\sqrt{Y/n}}$$

then r.v. T has a t distribution (student's t distribution) with degrees of freedom (df) n , which is denoted as

$$T \sim t(n).$$

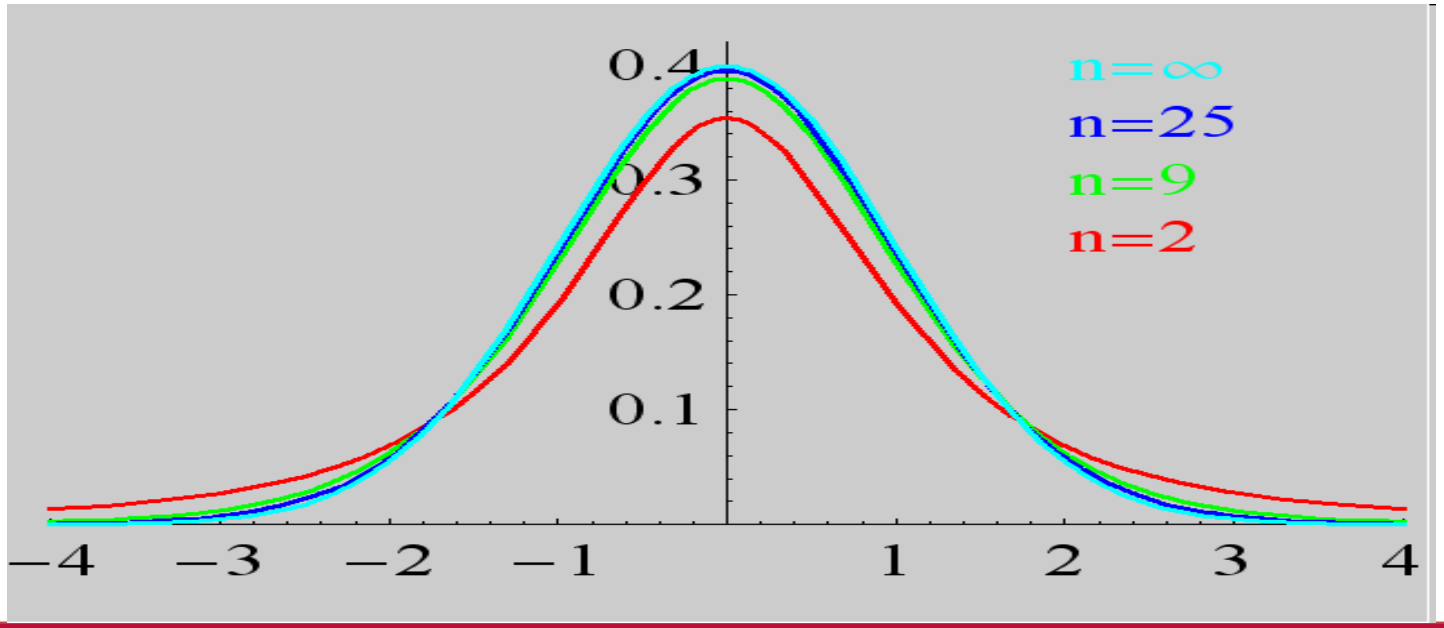
# William Sealey Gosset

**Finder of t distribution**

**Born: 13 June 1876 in Canterbury, England**
**Died: 16 Oct 1937 in Beaconsfield, England**

2) t(n)'s pdf

$$f(t)=\frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{\pi n}\,\Gamma\left(\frac{n}{2}\right)}\left(1+\frac{t^2}{n}\right)^{-\frac{n+1}{2}},\quad -\infty<t<+\infty$$



# pdf of t(n)' properties:

i. Bell shaped, symmetric about 0.

ii. More spread out than N(0, 1) curve.

iii. As $n \to \infty$ , $t(n) \to N(0,1)$. (Approximation is good when $n \geq 30$ )

iv. As n increases, the tails become thinner.

The tails of the t-distribution can approach zero much more slowly than those of the normal distribution which is called more heavy-tailed than the normal distribution.

# $t$(n)′ s properties:

1) If r.v. T~t(n) , then when n>2, we have

$$E(T) = 0 \qquad\qquad Var(T) = \frac{n}{n-2}$$

2) If r.v. T~t(n) , p(t) is T's pdf, then

$$\lim_{n\to\infty} f(t) = \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}}$$

this result means $n \to \infty$

$$t(n) \to N(0,1).$$

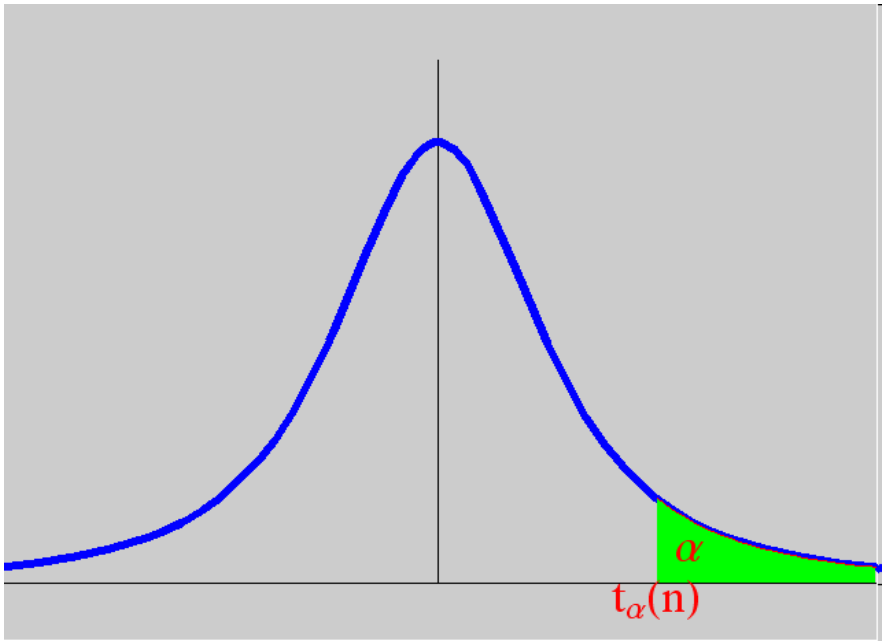
# Upper quantiles of t distribution

Definition: The value $t_{\alpha}^{2}(n)$ is the upper α percentile of the t distribution with degree n , such that α($\mathbf{0} < \boldsymbol{\alpha} < \mathbf{1}$) of the probability (or α of the area) lies to the right of this value. That is

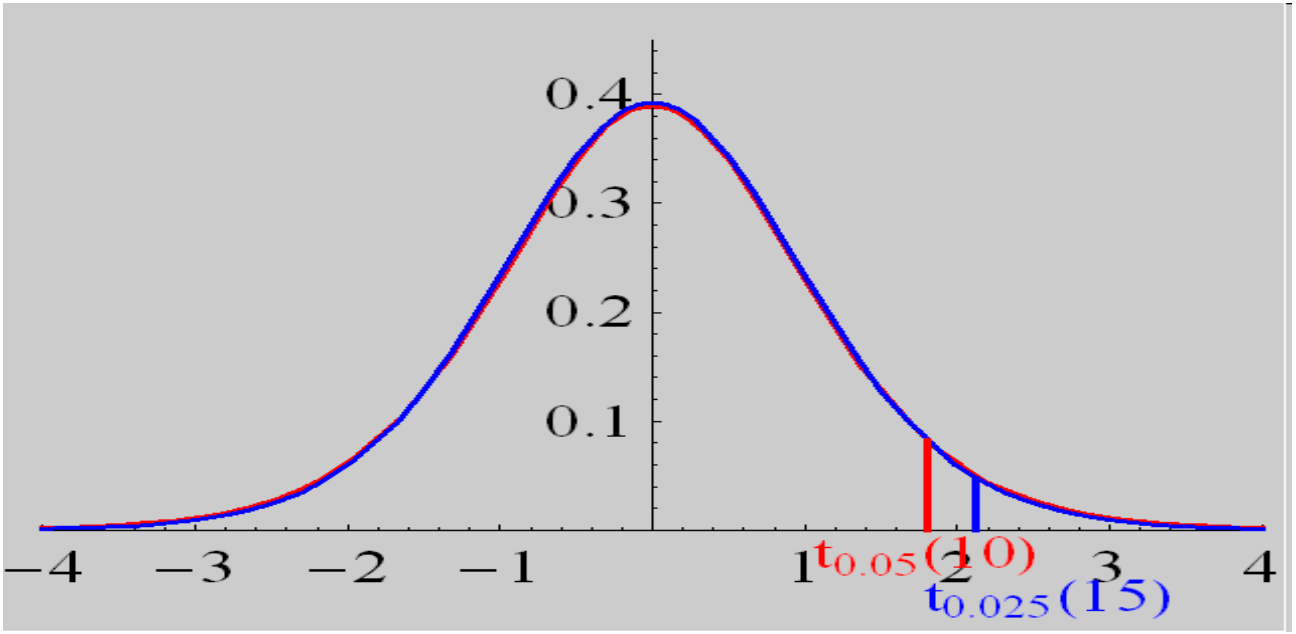
$$P\{t > t_{\alpha}^{2}(n)\} = \alpha$$

$$\boldsymbol{t_{1-\alpha}(n) = -t_{\alpha}(n).}$$

$$n > 45, \ t_{\alpha}(n) \approx u_{\alpha}.$$



Look up the quantile table of t distribution,




$$t_{0.05}(10) = 1.8125, \qquad t_{0.025}(15) = 2.1315$$

# T Distribution Table

| $n$ | $\alpha$ =0.25 | 0.10 | 0.05 | 0.025 | 0.01 | 0.005 |
|---|---|---|---|---|---|---|
| 1 | 1.0000 | 3.0777 | 6.3138 | 12.7062 | 31.8207 | 63.6574 |
| 2 | 0.8165 | 1.8856 | 2.9200 | 4.3027 | 6.9646 | 9.9248 |
| 3 | 0.7649 | 1.6377 | 2.3534 | 3.1824 | 4.5407 | 5.8409 |
| 4 | 0.7407 | 1.5332 | 2.1318 | 2.7764 | 3.7469 | 4.6041 |
| 5 | 0.7267 | 1.4759 | 2.0150 | 2.5706 | 3.3649 | 4.0322 |
| 6 | 0.7176 | 1.4398 | 1.9432 | [illegible] | [illegible]427 | 3.7074 |
| 7 | 0.7111 | 1.4149 | 1.8946 | [illegible] | [illegible]980 | 3.4995 |
| 8 | 0.7064 | 1.3968 | 1.8595 | [illegible] | [illegible]065 | 3.3554 |
| 9 | 0.7027 | 1.3830 | 1.8331 | 2.2622 | 2.8214 | 3.2498 |
| 10 | 0.6998 | 1.3722 | 1.8125 | 2.2281 | 2.7638 | 3.1693 |
| 11 | 0.6974 | 1.3634 | 1.7959 | 2.2010 | 2.7181 | 3.1058 |
| 12 | 0.6955 | 1.3562 | 1.7823 | 2.1788 | 2.6810 | 3.0545 |
| 13 | 0.6938 | 1.3502 | 1.7709 | 2.1604 | 2.6503 | 3.0123 |
| 14 | 0.6924 | 1.3450 | 1.7613 | 2.1448 | 2.6245 | 2.9768 |
| 15 | 0.6912 | 1.3406 | 1.7531 | 2.1315 | 2.6025 | 2.9467 |
| 16 | 0.6901 | 1.3368 | 1.7459 | 2.1199 | 2.5835 | 2.9208 |

**1.8125**

# T Distribution Table

| $n$ | $\alpha$ =0.25 | 0.10 | 0.05 | 0.025 | 0.01 | 0.005 |
|---|---|---|---|---|---|---|
| 1 | 1.0000 | 3.0777 | 6.3138 | 12.7062 | 31.8207 | 63.6574 |
| 2 | 0.8165 | 1.8856 | 2.9200 | 4.3027 | 6.9646 | 9.9248 |
| 3 | 0.7649 | 1.6377 | 2.3534 | 3.1824 | 4.5407 | 5.8409 |
| 4 | 0.7407 | 1.5332 | 2.1318 | 2.7764 | 3.7469 | 4.6041 |
| 5 | 0.7267 | 1.4759 | 2.0150 | 2.5706 | 3.3649 | 4.0322 |
| 6 | 0.7176 | 1.4398 | 1.9432 | 2.4469 | 3.1427 | 3.7074 |
| 7 | 0.7111 | 1.4149 | 1.8946 | 2.3646 | 2.9980 | 3.4995 |
| 8 | 0.7064 | 1.3968 | 1.8595 | 2.3060 | 2.8965 | 3.3554 |
| 9 | 0.7027 | 1.3830 | 1.8331 | 2.2622 | 2.8214 | 3.2498 |
| 10 | 0.6998 | 1.37 | | 2.2281 | 2.7638 | 3.1693 |
| 11 | 0.6974 | 1.36 | | 2.2010 | 2.7181 | 3.1058 |
| 12 | 0.6955 | 1.35 | | 2.1788 | 2.6810 | 3.0545 |
| 13 | 0.6938 | 1.3502 | 1.7709 | 2.1604 | 2.6503 | 3.0123 |
| 14 | 0.6924 | 1.3450 | 1.7613 | 2.1448 | 2.6245 | 2.9768 |
| 15 | 0.6912 | 1.3406 | 1.7531 | 2.1315 | 2.6025 | 2.9467 |
| 16 | 0.6901 | 1.3368 | 1.7459 | 2.1199 | 2.5835 | 2.9208 |

**2.1315**

# The F distribution

1） Definition

If $X \sim \chi^2(n_1),\ Y \sim \chi^2(n_2), X, Y$ are independent,

then r.v.

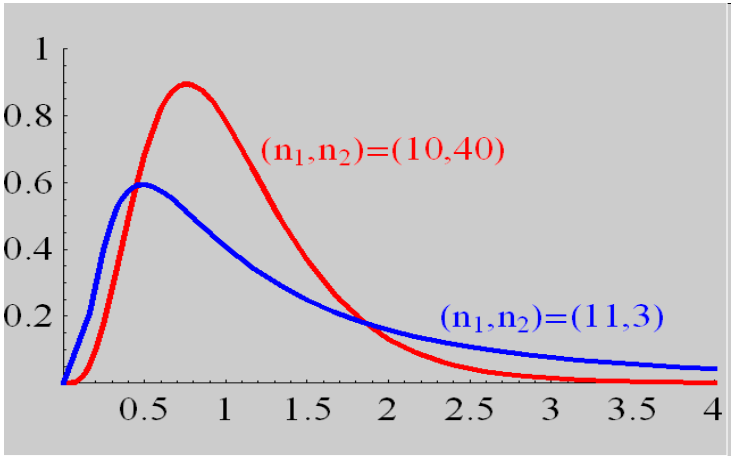
$$F = \frac{X / n_1}{Y / n_2}$$

is distributed as F with degrees of freedom $n_1, n_2$, which is denoted as

$$F \sim F(n_1, n_2).$$

2) pdf of $F(n_1, n_2)$ distribution

$$f(y) = \begin{cases} \dfrac{\Gamma\left(\dfrac{n_1+n_2}{2}\right)\left(\dfrac{n_1}{n_2}\right)^{\frac{n_1}{2}} y^{\frac{n_1}{2}-1}}{\Gamma\left(\dfrac{n_1}{2}\right)\Gamma\left(\dfrac{n_2}{2}\right)\left[1+\left(\dfrac{n_1 y}{n_2}\right)\right]^{\frac{n_1+n_2}{2}}}, & y > 0 \\ 0, & otherwise \end{cases}$$



# F distribution’s properties

(1) $$F \sim F(n_1, n_2) \Rightarrow \frac{1}{F} \sim F(n_2, n_1).$$

(2) $$E(F) = \frac{n_2}{n_2 - 2}, \quad (n_2 > 2),$$

$$Var(F) = \frac{2n_2^2(n_1 + n_2 - 2)}{n_1(n_2 - 2)^2(n_2 - 4)}, \quad (n_2 > 4)$$

(3) $$T \sim t(n) \Rightarrow \quad T^2 \sim F(1, n).$$

**(4)** If $F \sim F(n_1, n_2)$, when $n_2 > 4$, for any x,

$$\lim_{n_1 \to \infty} P\{\frac{F - E(F)}{\sqrt{Var(F)}} \le x\} = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

That's means, when $n_1 \longrightarrow \infty$

$$\frac{F - E(F)}{\sqrt{Var(F)}} \to N(0,1)$$

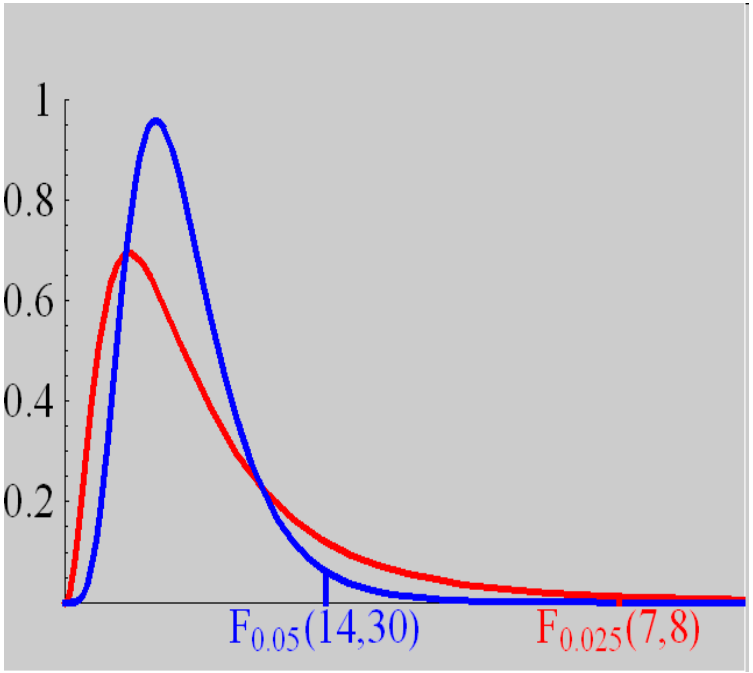
# Upper quantiles of F distribution

Definition: The value $F_\alpha(n_1, n_2)$ is the upper α percentile of the distribution $F(n_1, n_2)$ such that $\alpha(\mathbf{0} < \boldsymbol{\alpha} < \mathbf{1})$ of the probability (or α of the area) lies to the right of this value. That is

$$P\{F > F_\alpha(n_1, n_2)\} = \alpha$$

$$F_{0.025}(8,7) = 4.9$$

$$F_{0.05}(30,14) = 2.31$$



# F Distribution Table

## $\alpha = 0.025$

| $n_2$ \ $n_1$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 12 | 15 | 20 | 24 | 30 | 40 | 120 | ∞ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 647.8 | 799.5 | 864.2 | 899.6 | 921.8 | 937.1 | 948.2 | 956.7 | 963.3 | 968.6 | 976.7 | 984.9 | 993.1 | 997.2 | 1001 | 1006 | 1014 | 1018 |
| 2 | 38.51 | 39.00 | 39.17 | 39.25 | 39.30 | 39.33 | 39.36 | 39.37 | 39.39 | 39.40 | 39.41 | 39.43 | 39.45 | 39.46 | 39.46 | 39.47 | 39.49 | 39.50 |
| 3 | 17.44 | 16.04 | 15.44 | 15.10 | 14.88 | 14.73 | 14.62 | 14.54 | 14.47 | 14.42 | 14.34 | 14.25 | 14.17 | 14.12 | 14.08 | 14.04 | 13.95 | 13.90 |
| 4 | 12.22 | 10.65 | 9.98 | 9.60 | 9.36 | 9.20 | 9.07 | 8.98 | 8.90 | 8.84 | 8.75 | 8.66 | 8.56 | 8.51 | 8.46 | 8.41 | 8.31 | 8.26 |
| 5 | 10.01 | 8.43 | 7.76 | 7.39 | 7.15 | 6.98 | 6.85 | 6.76 | 6.68 | 6.62 | 6.52 | 6.43 | 6.33 | 6.28 | 6.23 | 6.18 | 6.07 | 6.02 |
| 6 | 8.81 | 7.26 | 6.60 | 6.23 | 5.99 | 5.82 | 5.70 | 5.60 | 5.52 | 5.46 | 5.37 | 5.27 | 5.17 | 5.12 | 5.07 | 5.01 | 4.90 | 4.85 |
| 7 | 8.07 | 6.54 | 5.89 | 5.52 | 5.29 | 5.12 | 4.99 | 4.90 | 4.82 | 4.76 | 4.67 | 4.57 | 4.47 | 4.42 | 4.36 | 4.31 | 4.20 | 4.14 |
| 8 | 7.57 | 6.06 | 5.42 | 5.50 | 4.82 | 4.65 | 4.53 | [illegible] | 4.36 | 4.30 | 4.20 | 4.10 | 4.00 | 3.59 | 3.89 | 3.84 | 3.73 | 3.67 |
| 9 | 7.21 | 5.71 | 5.08 | 4.72 | 4.48 | 4.23 | 4.20 | [illegible] | 4.03 | 3.96 | 3.87 | 3.77 | 3.67 | 3.61 | 3.56 | 3.51 | 3.39 | 3.33 |
| 10 | 6.94 | 5.46 | 4.83 | 4.47 | 4.24 | 4.07 | 3.95 | [illegible] | [illegible] | 3.72 | 3.62 | 3.52 | 3.42 | 3.37 | 3.31 | 3.26 | 3.14 | 3.08 |
| 11 | 6.72 | 5.26 | 4.63 | 4.28 | 4.04 | 3.88 | 3.76 | [illegible] | [illegible] | 3.53 | 3.43 | 3.33 | 3.23 | 3.17 | 3.12 | 3.06 | 2.94 | 2.88 |
| 12 | 6.55 | 5.10 | 4.47 | 4.12 | 3.89 | 3.73 | 3.61 | [illegible] | [illegible] | [illegible] | [illegible] | 3.18 | 3.07 | 3.02 | 2.96 | 2.91 | 2.79 | 2.72 |
| 13 | 6.41 | 4.97 | 4.35 | 4.00 | 3.77 | 3.60 | 3.48 | [illegible] | [illegible] | [illegible] | [illegible] | 3.05 | 2.95 | 2.89 | 2.84 | 2.78 | 2.66 | 2.60 |
| 14 | 6.30 | 4.86 | 4.24 | 3.89 | 3.66 | 3.50 | 3.38 | [illegible] | [illegible] | [illegible] | [illegible] | 2.95 | 2.84 | 2.79 | 2.73 | 2.67 | 2.55 | 2.49 |
| 15 | 6.20 | 4.77 | 4.15 | 3.80 | 3.58 | 3.41 | 3.29 | [illegible] | [illegible] | [illegible] | [illegible] | 2.86 | 2.76 | 2.70 | 2.64 | 2.59 | 2.46 | 2.40 |
| 16 | 6.12 | 4.69 | 4.08 | 3.73 | 3.50 | 3.34 | 3.22 | [illegible] | [illegible] | [illegible] | [illegible] | 2.79 | 2.68 | 2.63 | 2.57 | 2.51 | 2.38 | 2.32 |
| 17 | 6.04 | 4.62 | 4.01 | 3.66 | 3.44 | 3.28 | 3.16 | 3.06 | 2.98 | 2.92 | 2.82 | 2.72 | 2.62 | 2.56 | 2.50 | 2.44 | 2.32 | 2.25 |
| 18 | 5.95 | 4.56 | 3.95 | 3.61 | 3.38 | 3.22 | 3.10 | 3.01 | 2.93 | 2.87 | 2.77 | 2.67 | 2.56 | 2.50 | 2.44 | 2.38 | 2.26 | 2.19 |
| 19 | 5.92 | 4.51 | 3.90 | 3.56 | 3.33 | 3.17 | 3.05 | 2.96 | 2.88 | 2.82 | 2.72 | 2.62 | 2.51 | 2.45 | 2.39 | 2.33 | 2.20 | 2.13 |

**4.90**

# F Distribution Table

## $\alpha = 0.05$

| $n_2$ \ $n_1$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 15 | 20 | 24 | 30 | 40 | 60 | 120 | ∞ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 161.4 | 199.5 | 215.7 | 224.6 | 230.2 | 234.0 | 236.8 | 238.9 | 24.05 | 241.9 | 245.9 | 248.0 | 249.1 | 250.1 | 151.1 | 252.2 | 253.3 | 254.3 |
| 2 | 18.51 | 19.00 | 19.16 | 19.25 | 19.30 | 19.33 | 19.35 | 19.37 | 19.38 | 19.40 | 19.43 | 19.45 | 19.45 | 19.46 | 19.47 | 19.48 | 19.49 | 19.50 |
| 3 | 10.13 | 9.55 | 9.28 | 9.12 | 9.01 | 8.94 | 8.89 | 8.85 | 8.81 | 8.79 | 8.70 | 8.66 | 8.64 | 8.62 | 8.59 | 8.57 | 8.55 | 8.53 |
| 4 | 7.71 | 6.94 | 6.59 | 6.39 | 6.26 | 6.16 | 6.09 | 6.04 | 6.00 | 5.96 | 5.86 | 5.80 | 5.77 | 5.75 | 5.72 | 5.69 | 5.66 | 5.63 |
| 5 | 6.61 | 5.79 | 5.41 | 5.19 | 5.05 | 4.95 | 4.88 | 4.82 | 4.77 | 4.74 | 4.62 | 4.56 | 4.53 | 4.50 | 4.46 | 4.43 | 4.40 | 4.36 |
| 6 | 5.99 | 5.14 | 4.76 | 4.53 | 4.39 | 4.28 | 4.21 | 4.15 | 4.10 | 4.06 | 3.94 | 3.87 | 3.84 | 3.81 | 3.77 | 3.74 | 3.70 | 3.67 |
| 7 | 5.59 | 4.74 | 4.35 | 4.12 | 3.97 | 3.87 | 3.79 | 3.73 | 3.68 | 3.64 | 2.51 | 3.44 | 3.41 | 3.38 | 3.34 | 3.30 | 3.27 | 3.23 |
| 8 | 5.32 | 4.46 | 4.07 | 3.84 | 3.69 | 3.58 | 3.50 | 3.44 | 3.39 | 3.35 | 3.22 | 3.15 | 3.12 | 3.08 | 3.04 | 3.01 | 2.97 | 2.93 |
| 9 | 3.12 | 4.26 | 3.81 | 3.63 | 3.48 | 3.37 | 3.29 | 3.23 | 3.18 | 3.14 | 3.01 | 2.94 | 2.90 | 2.86 | 2.83 | 2.79 | 2.75 | 2.71 |
| 10 | 4.96 | 4.10 | 3.71 | 3.48 | 3.33 | 3.22 | 3.14 | 3.07 | 3.02 | 2.98 | 2.85 | 2.77 | 2.74 | 2.70 | 2.66 | 2.62 | 2.58 | 2.54 |
| 11 | 4.84 | 3.98 | 3.59 | 3.36 | 3.20 | 3.09 | 3.01 | 2 | | | | | 2.61 | 2.57 | 2.53 | 2.49 | 2.45 | 2.40 |
| 12 | 4.75 | 3.89 | 3.49 | 3.26 | 3.11 | 3.00 | 2.91 | 2 | | | | | 2.51 | 5.47 | 2.43 | 2.38 | 2.34 | 2.30 |
| 13 | 4.67 | 3.81 | 3.41 | 3.18 | 3.03 | 2.92 | 2.83 | 2 | | | | | 2.42 | 2.38 | 2.34 | 2.30 | 2.25 | 2.21 |
| 14 | 4.60 | 3.74 | 3.34 | 3.11 | 2.96 | 2.85 | 2.76 | 2 | | | | | | 2.31 | 2.27 | 2.22 | 2.18 | 2.13 |
| 15 | 4.54 | 3.68 | 3.29 | 3.06 | 2.90 | 2.79 | 2.71 | 2.6 | | | | 3 | 2.29 | 2.25 | 2.20 | 2.16 | 2.11 | 2.07 |
| 16 | 4.49 | 3.63 | 3.24 | 3.01 | 2.85 | 2.74 | 2.66 | 2.59 | 2.54 | 2.49 | 2.35 | 2.28 | 2.24 | 2.19 | 2.15 | 2.11 | 2.06 | 2.01 |
| 17 | 4.45 | 3.59 | 3.20 | 2.96 | 2.81 | 2.70 | 2.61 | 2.55 | 2.49 | 2.45 | 2.31 | 2.23 | 2.19 | 2.15 | 2.10 | 2.06 | 2.01 | 1.96 |
| 18 | 4.41 | 5.55 | 3.16 | 2.93 | 2.77 | 2.66 | 2.58 | 2.51 | 2.46 | 2.41 | 2.27 | 2.19 | 2.15 | 2.11 | 2.06 | 2.02 | 1.97 | 1.92 |
| 19 | 4.38 | 3.52 | 3.13 | 2.90 | 2.74 | 2.63 | 2.54 | 2.48 | 2.42 | 2.38 | 2.23 | 2.16 | 2.11 | 2.07 | 2.03 | 1.98 | 1.93 | 1.88 |

**2.31**

Property: quantiles of F distribution

$$F_{1-\alpha}(n_1, n_2) = \frac{1}{F_{\alpha}(n_2, n_1)}$$

example:

$$F_{0.95}(12,9) = \frac{1}{F_{0.05}(9,12)} = \frac{1}{2.8} = 0.375$$

# Percentiles

One measure of variation is the sample range, which is the difference between the maximum and the minimum observations.

Detailed information about the measure of location and measure of variation can be obtained using ordered data.

Percent ranks of a data set are called percentiles.

# Percentiles

Sample 100p-th percentile: value such that at least 100p% of the observations are at or below this value, and at least $100(1-p)\%$ are at or above this value.

Quartiles are frequently-used percentiles. Quartiles are the points of division into quarters.

First quartile (Q1): 25th percentile

Second quartile (Q2): 50th percentile, median

Third quartile (Q3): 75th percentile